

Unit 1, Lesson 1: Rewriting Rational Numbers in Equivalent Forms

Benchmark

MA.7.NSO.1.2 Rewrite rational numbers in different but equivalent forms including fractions, mixed numbers, repeating decimals, and percentages to solve mathematical and real-world problems.

Learning Target

☐ I can rewrite rational numbers in different but equivalent forms.

1.1.1 Warm-Up: Day at Work: Senior Designer

Nik Brensnick is a senior product designer. She is quoted as saying, "A big part of working and being successful in the workplace has to do with cultivating relationships with other people."

1. Why do you think Nik has this belief about being successful in the workplace?



1.1.2 Activity: Match It Up!

1. Work with your partner to match the equivalent rational numbers expressed in different forms. Record each matching set in the table.

Fraction	Decimal	Percent

2. Explain your method for matching the equivalent rational forms.

1.1.3 Guided Practice: Repeating Decimals



A *rational number* is a real number that can be expressed as a ratio of two integers.

Rational numbers can be written as a fraction, $\frac{a}{b}$, where a and b are integers and $b \neq 0$, or as a decimal. In decimal form, rational numbers either terminate or repeat.

1. Consider the fractions in the table shown.
- a. Complete the table by rewriting each fraction into its equivalent form. If a decimal repeats digits in a pattern, use an ellipse (...) to indicate that the pattern continues.

Fraction Form	Decimal Form	Terminating or Repeating?
$\frac{1}{6}$		☐ terminating ☐ repeating
$\frac{5}{8}$		☐ terminating ☐ repeating
$\frac{4}{9}$		☐ terminating ☐ repeating

- b. Based on your work in Part A, explain whether it is possible for a repeating decimal to ever terminate.



A number with a decimal that never ends, but has a pattern that repeats infinitely, is called a **repeating decimal**.

Repeating decimals are written with a bar over the digits that repeat.

For example, the fraction $\frac{17}{6}$ is expressed as $2.8\bar{3}$ when written as a decimal because the 3 is the digit that repeats infinitely.

2. Dulce divided 73 by 99 using a calculator to rewrite $\frac{73}{99}$ in decimal form. She found the resulting pattern 0.737373 ...

Write $\frac{73}{99}$ in its equivalent decimal form.

1.1.4 Guided Instruction: Rewriting Repeating Decimals as Fractions

1. Consider the repeating decimal $0.4\overline{1}$.

a. If $x = 0.4\overline{1}$, complete each equation.

$$x = 0.4\overline{1}$$

$$10x = \underline{\hspace{2cm}}$$

$$100x = \underline{\hspace{2cm}}$$

b. Which two values, when subtracted, would result in the elimination of the repeating portion of the decimal?

c. Use two of the relationships represented in the equations in Part A to complete the steps to find the fraction equivalent of $0.4\overline{1}$.

$$\underline{\hspace{1cm}} - \underline{\hspace{1cm}} = \underline{\hspace{1cm}} - \underline{\hspace{1cm}}$$

$$\underline{\hspace{1cm}}x = \underline{\hspace{1cm}}$$

$$x = \frac{\boxed{}}{\boxed{}}$$

d. Rewrite the solution as a fraction in the simplest form, if possible.

1.1.5 Try It: Equivalent Forms

1. Complete the table.

Fraction	Decimal	Percent
$\frac{15}{99}$		
		32%
	$0.\overline{72}$	
$\frac{5}{16}$		
		$88.\overline{8}\%$
	$0.5\overline{2}$	
$\frac{1}{12}$		

1.1.6 Wrap-Up: Reflection

1. Check all the statements that apply to your understanding.
 - ☐ I understand how to rewrite rational numbers in different but equivalent forms.
 - ☐ I need support on how to rewrite rational numbers in different but equivalent forms.
 - ☐ I understand how to convert a repeating decimal to a fraction.
 - ☐ I need support on how to convert a repeating decimal to a fraction.